

Hayden Supple

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PROFESSIONAL EXPERIENCE

Engineering Intern @ Bedrock AI

March – September 2025

Palo Alto, California

- **Designed and built** a V1 **portable, self-hosted AI hardware**, designing custom sheet aluminum case, dual **custom** OLED displays (13.3" & 5.5"), component **specification**, and **assembly**.
- **Applied GD&T and DFM** to ensure proper fit, alignment, and clearances across the enclosure, display mounts, and internal hardware assemblies.
- **Developed** core **software** features, including **on-prem AI Hosting**, tunnel porting, personal vector database, and an AI-powered, Cursor-Equivalent, document editor.

Vehicle Systems Engineer @ Purdue Autonomous Racing

January 2025 – Present

West Lafayette, Indiana

- **Built** app for live ROS2 **telemetry** across vehicle subsystems at speeds exceeding **150 MPH**.
- **Engineered** GPS-based **live mapping** with coordinate-flattening for **precise** race visualization.

ENGINEERING PROJECTS

Autonomous "UnLate" Skateboard + Swift App Integration

May – October 2025

- **Engineered a fully autonomous electric skateboard**
 - Designing dual BLDC motor drive powertrain
 - CNC and Hand-machined 6061 Custom Designed Aluminum trucks and precision rack-and-pinion steering mechanism.
- Built a **12s3p 540 Wh battery** with cell balancing, and custom case.
- Integrated embedded controls with a self made Swift guidance app using on-device YOLO object detection and GPS-based routing.

NMHS - Electronic Shelf Labeling

September 2025– Present

- **Designed ML platform** using YOLO pattern detection and Wifi Connection for data distribution.
 - Direct adopt technology for ease of system transfer and adoption from analog.
- **Developed** shelf-integrated Electronic Tag Systems
 - DFM Principles to be mass produced
 - Universal configuration for Retail Stores
- **Streamlined** manual shelf labeling, saving **10+ labor hours/week** for **50%** competitor pricing.

EDUCATION

B.S Mechanical Engineering, Purdue University

August 2024 – May 2027

West Lafayette, Indiana

3.89 / 4.0 GPA

LEADERSHIP EXPERIENCE

Governmental Relations Executive Director @ Purdue Student Gov

November 2024 – Present

West Lafayette, Indiana

- **Lobbied** for Indiana Lifeline Bill (SB74), achieving unanimous 9–0 **passage** in state senate.
- **Represented** Purdue at Big Ten on the Hill, advocating on issues affecting **50,000+** students in meetings with U.S. senators.

TECHNICAL SKILLS

Mechanical & Electronics: CAD (Fusion 360, NX), CNC, Motor/Powertrain Design, Battery Systems, Microcontrollers, Motor Drivers, On-Board Computing, FEA (Ansys)

Programming & Software: C++, Python, JavaScript, React, Swift, ROS2, CoreML, YOLO, MATLAB